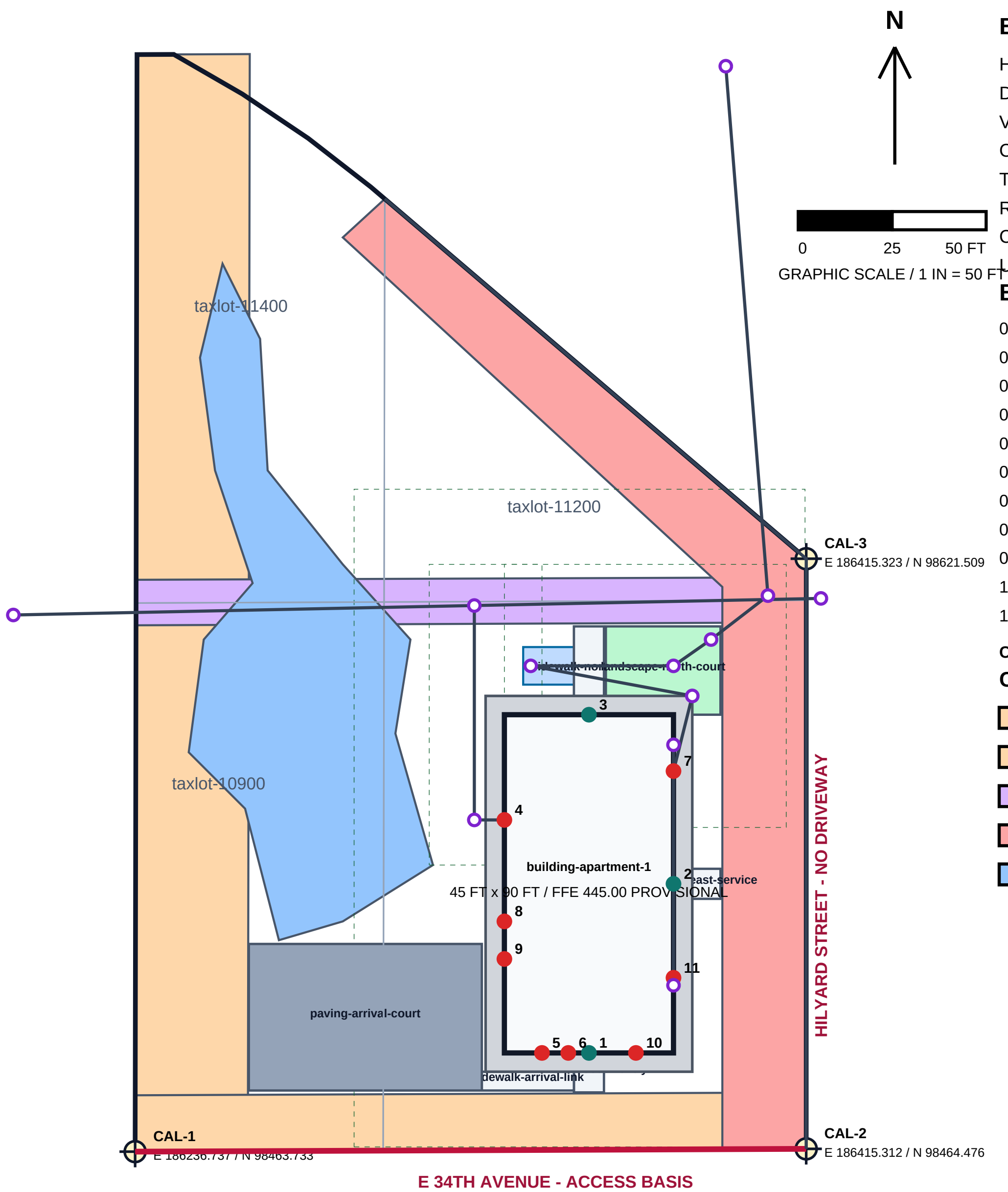
MODEL STUDIO - HILYARD APARTMENT CIVIL PLAN TEST SITE LAYOUT / BUILDING INTERFACES

Product-development test data only - no survey, engineering, capacity, approval, or code-compliance claim



BASIS / CONTROL

Horizontal: EPSG:6823, international feet
DRAWING GRID ORIGIN: E 186225.00 / N 98450.00
Vertical: NAVD88 FT; project benchmark UNKNOWN
City contours: NGVD29 +3.698 FT via NOAA VDatum
Taxlots: City GIS reference-derived; NOT A SURVEY
Replace taxlot geometry first when a boundary survey arrives
Constraints: dimensioned/digitized reference geometry
Utility routing is on discipline sheets; capacity/approval unknown

BUILDING INTERFACES - PERMANENT TERMINAL IDS

01 entry-south-primary [pedestrian]
02 entry-east-service [service]
03 entry-north-pedestrian [pedestrian]
04 penetration-sanitary [sanitary]
05 penetration-domestic-water [domestic_water]
06 penetration-fire-water [fire_water]
07 penetration-roof-drainage [roof_drainage]
08 penetration-electric [electric]
09 penetration-telecom-fiber [telecom_fiber]
10 penetration-gas [gas]
11 penetration-site-lighting [site_lighting]

CAL-1/2/3: REFERENCE-SCALE TEST CONTROL — NOT FOR STAKING

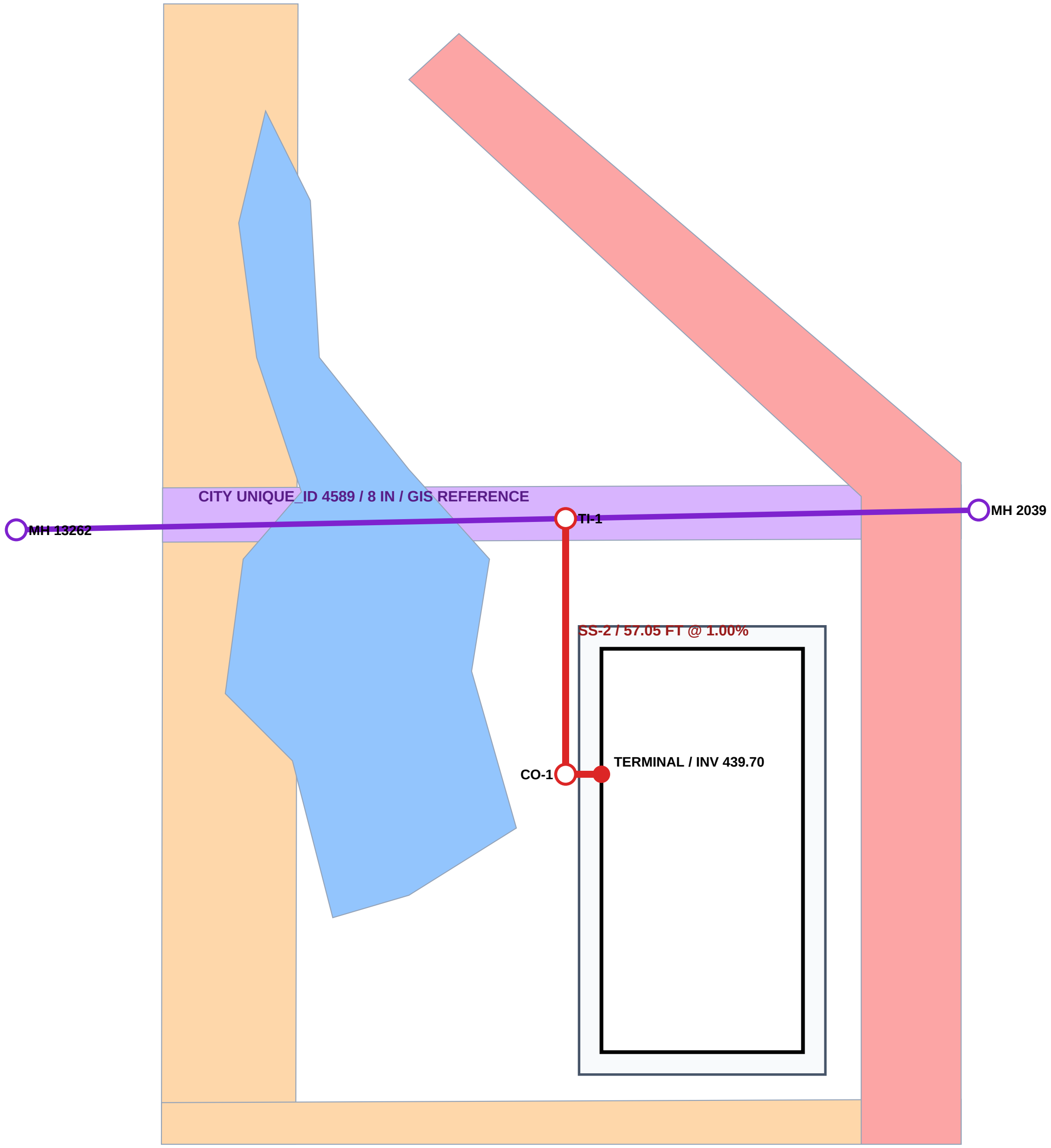
CONSTRAINT LEGEND

- constraint-eweb-west
- constraint-eweb-south
- constraint-sanitary-easement
- constraint-row-dedication
- constraint-wetland

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

MODEL STUDIO - HILYARD APARTMENT CIVIL PLAN TEST SANITARY SEWER PLAN

Reference GIS context plus reviewed-assumption fictional service; no survey, capacity, tie-in approval, or final grading claim



SANITARY BASIS

City main: UNIQUE_ID 4589
8 IN / folded-form liner / reference GIS
GIS inverts: 438.87 -> 437.47 NAVD88
Proposed service: 6 IN PVC SDR35
SS-1: 8.00 FT @ 1.00%
SS-2: 57.05 FT @ 1.00%
TI-1 incoming INV 439.05 (assumed)
Public main INV at TI-1: 438.07 (interpolated)
Minimum cover checked: 4.00 FT
All proposed cover <5 FT: shallow review warning
Tie-in approval and capacity: UNKNOWN
Surface/FFE: PROVISIONAL; not survey control

FIELD WORKFLOW

1 Pothole/survey existing main
2 Obtain City tie and capacity acceptance
3 Verify FFE, terminal, and finished grade
4 Stake line/grade; inspect bedding
5 Test, inspect, backfill/compact
6 Capture as-built coordinates/inverts

STABLE ASSET IDS

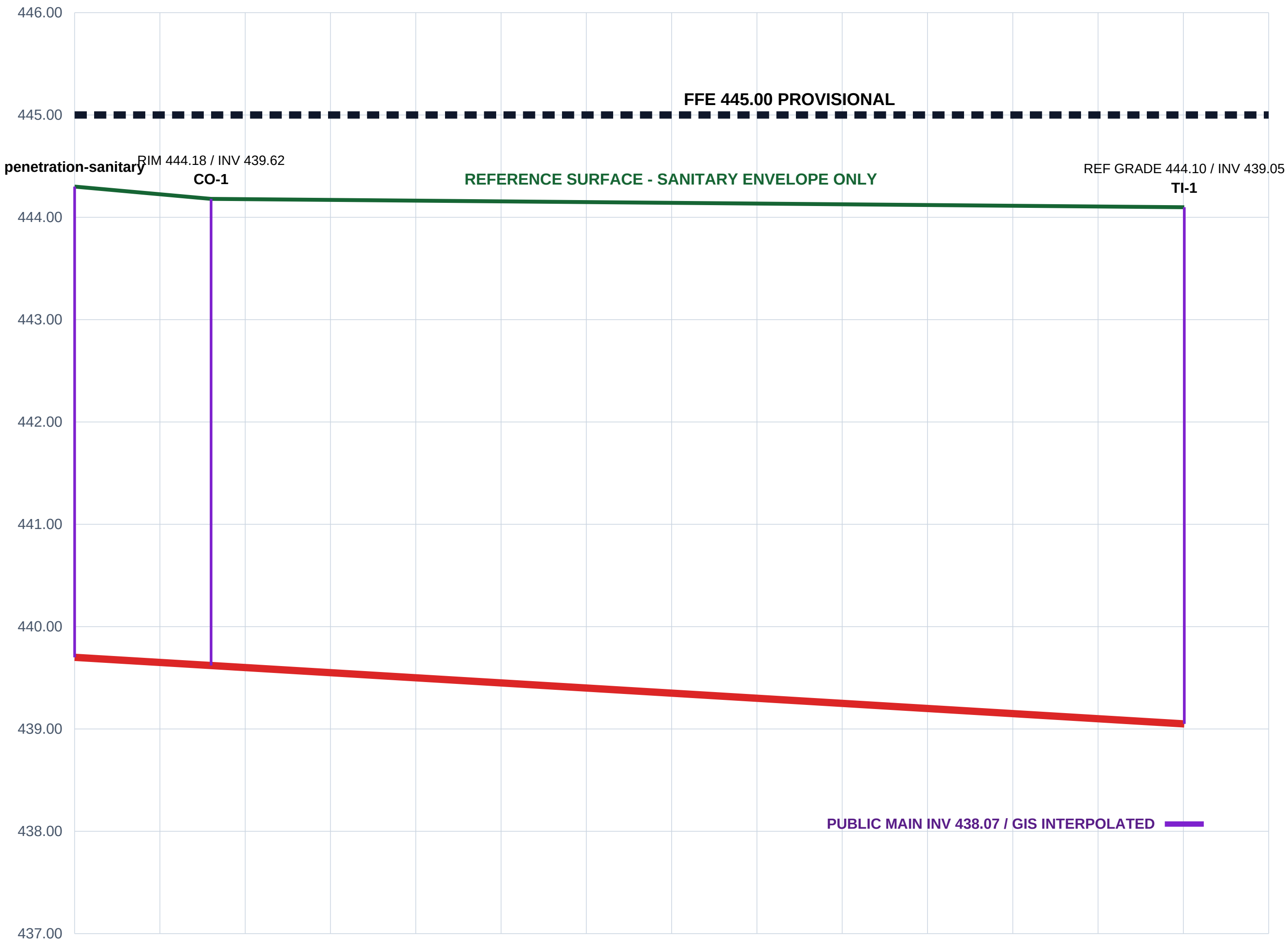
sanitary-service-seg-01
sanitary-service-seg-02
sanitary-cleanout-01
sanitary-tie-in-01
sanitary-public-main-4589-upstream
sanitary-public-main-4589-downstream

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

MODEL STUDIO - HILYARD APARTMENT CIVIL PLAN TEST SANITARY SEWER PROFILE

Stations follow penetration-sanitary to TI-1; all proposed vertical values are provisional test assumptions



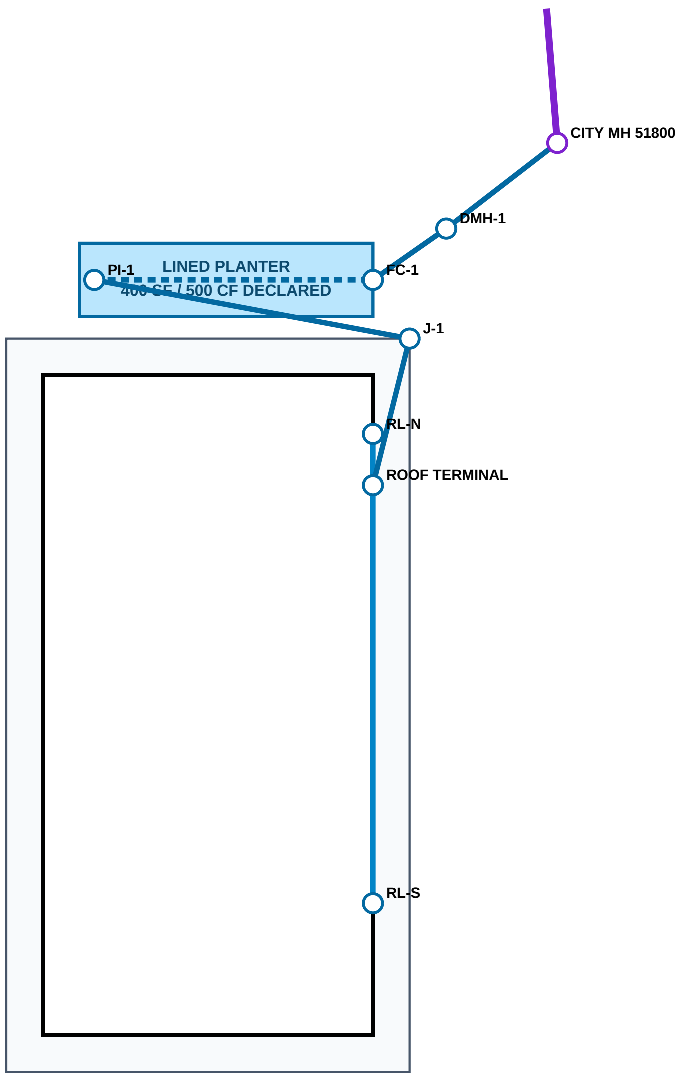
SEGMENT	STATIONS	LENGTH	SLOPE	INVERTS NAVD88	COVER	MATERIAL
san-edge-service-01	0+00.00 - 0+08.00	8.00 FT	1.00%	439.70 -> 439.62	4.06-4.10 FT	6 IN PVC SDR35
san-edge-service-02	0+08.00 - 0+65.05	57.05 FT	1.00%	439.62 -> 439.05	4.06-4.55 FT	6 IN PVC SDR35

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

Tie, FFE, surface, capacity, and owner acceptance unresolved

MODEL STUDIO - HILYARD APARTMENT CIVIL PLAN TEST STORM / ROOF DRAINAGE PLAN

Roof-only fictional treatment/conveyance system; final grading, capacity, HGL, infiltration, outfall approval, and survey are unresolved



HYDROLOGIC / VERTICAL BASIS

Roof area: 4,050 SF / C = 1.00 (assumed)

WQ storm: 1.40 IN / screen volume 472.50 CF

10-year storm: 4.46 IN / unrouted volume 1,505.25 CF

Planter declared surface storage: 500.00 CF

Method: rainfall-volume screening only; NOT ROUTED

Reference surface / FFE 445.00: PROVISIONAL

Public main: City GIS UNIQUE_ID 4183 / 21 IN

MH 51800 rim 444.87 / public out INV 434.33

Capacity, HGL, tie/outfall approval: UNKNOWN

Infiltration/geotechnical suitability: UNKNOWN

ALIGNMENT / CONFLICT

Private roof collector: 6 IN PVC

Facility underdrain: 6 IN perforated PVC

Outlet/public connection: 10 IN PVC SDR35

DMH-1 external drop: 3.95 FT (assumed)

Storm below sanitary crossing clearance: 1.253 FT

Reviewed minimum clearance: 1.00 FT

Pothole/survey both utilities before real design

No catch basin/site runoff: final grading excluded

FIELD HOLD POINTS

1 Confirm survey/control and final grades

2 Verify roof drains/overflow with architect

3 Obtain geotechnical infiltration testing

4 Confirm public capacity/HGL and City tie approval

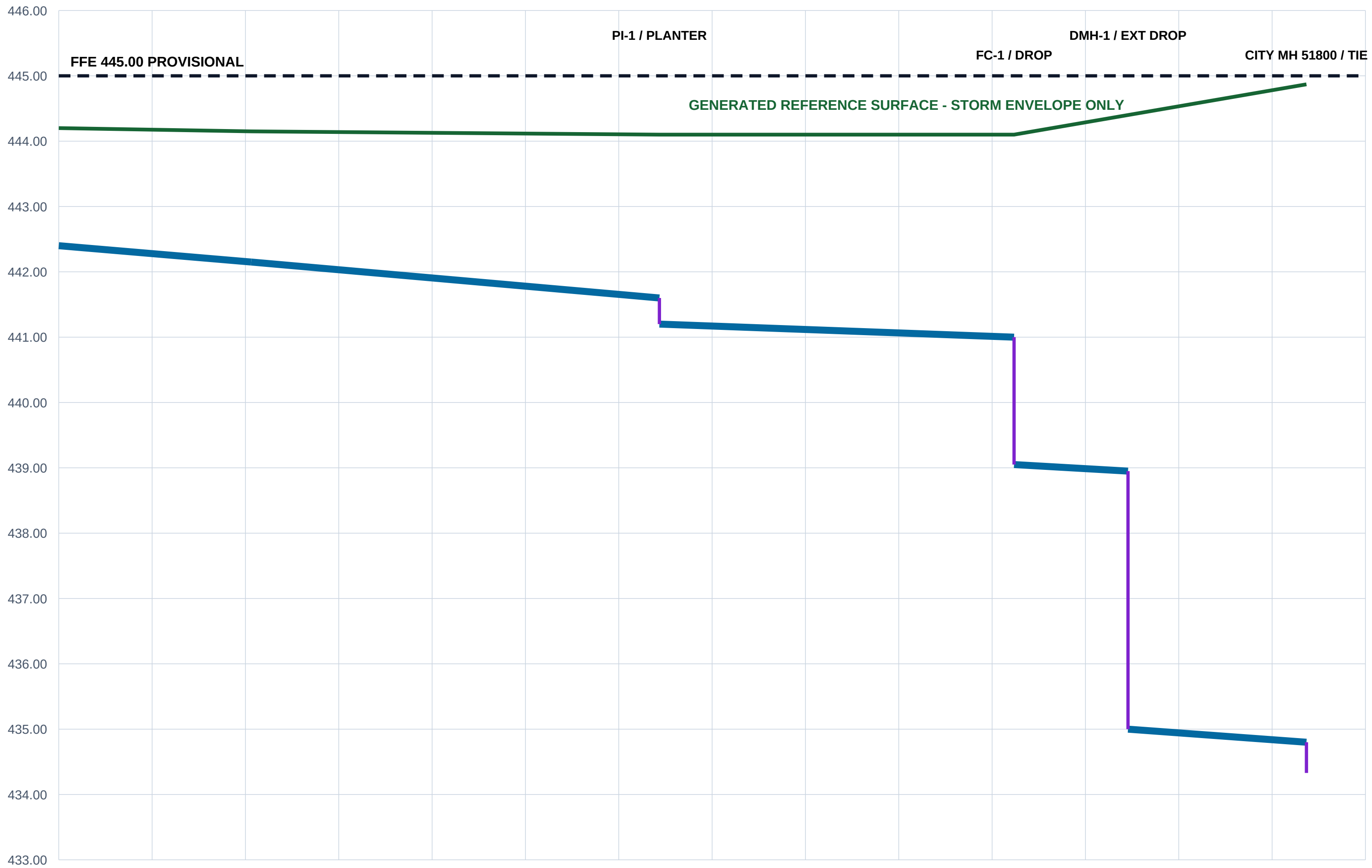
5 Pothole sanitary/storm crossing

6 Inspect liner, media, bedding, tests, as-builts

Reference City GIS is not survey, capacity, HGL, or connection authority

MODEL STUDIO - HILYARD APARTMENT CIVIL PLAN TEST STORM / ROOF DRAINAGE PROFILE

Stations run from permanent roof terminal through planter/flow control/drop manhole to assumed City tie; elevations are test-basis values



EDGE ID	STATIONS	LEN FT	SLOPE	INVERTS	COVER FT	MATERIAL
storm-edge-roof-lateral-01	0.00-20.62	20.62	1.21%	442.40->442.15	1.30-1.50	6 IN PVC
storm-edge-planter-inlet-01	20.62-64.35	43.74	1.26%	442.15->441.60	1.50-2.00	6 IN PVC
storm-edge-planter-underdrain-01	64.35-102.35	38.00	0.53%	441.20->441.00	2.40-2.60	6 IN PERFORATED PVC
storm-edge-outlet-01	102.35-114.56	12.21	0.82%	439.05->438.95	4.22-4.62	10 IN PVC SDR35
storm-edge-public-connection-01	114.56-133.68	19.12	1.05%	435.00->434.80	8.57-9.24	10 IN PVC SDR35

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

Positive gravity/cover checked against declared test basis; hydraulic capacity and HGL not established

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

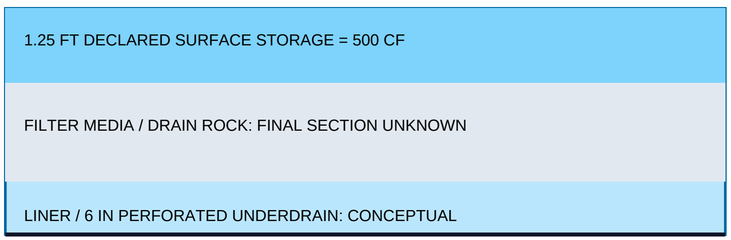
MODEL STUDIO - HILYARD APARTMENT CIVIL PLAN TEST STORM ASSET SCHEDULE / TEST DETAILS

Stable IDs, numeric test basis, field hold points, and explicit unknowns from the canonical semantic model

CONVEYANCE ASSET SCHEDULE

ASSET ID	TYPE	DIA	LEN	SLOPE	INVERTS NAVD88	STATUS
storm-edge-roof-lateral-01	private_roof_lateral	6 IN	20.62	1.21%	442.40->442.15	ASSUMED / GENERATED
storm-edge-planter-inlet-01	private_facility_lateral	6 IN	43.74	1.26%	442.15->441.60	ASSUMED / GENERATED
storm-edge-planter-underdrain-01	facility_underdrain	6 IN	38.00	0.53%	441.20->441.00	ASSUMED / GENERATED
storm-edge-outlet-01	private_storm_main	10 IN	12.21	0.82%	439.05->438.95	ASSUMED / GENERATED
storm-edge-public-connection-01	private_to_public_storm_connection	10 IN	19.12	1.05%	435.00->434.80	ASSUMED / GENERATED
roof-edge-leader-north	roof_leader	4 IN	7.00	1.43%	442.50->442.40	ASSUMED / GENERATED
roof-edge-leader-south	roof_leader	4 IN	57.00	1.05%	443.00->442.40	ASSUMED / GENERATED

LINED PLANTER TEST-BASIS DETAIL (NOT A CONSTRUCTION DETAIL)



REQUIRED REVIEW / HOLD POINTS

1. Survey boundary, structures, rims/inverts, final grades, and benchmark.
2. Confirm architectural roof drains, overflow paths, and plumbing sizes.
3. Perform geotechnical infiltration/groundwater testing; no infiltration credit now.
4. Route hydrographs and size treatment/flow control/flood overflow.
5. Obtain City capacity, HGL, tie-in/outfall, material, and detail acceptance.
6. Pothole and survey sanitary/storm crossing; maintain accepted separation.
7. Inspect bedding, backfill, compaction, liner/media, testing, and as-builts.

KNOWN-ANSWER TEST CHECKS

Roof leaders connect through permanent terminal penetration-roof-drainage.

Every proposed storm segment falls downstream and meets its declared minimum slope/cover.

FC-1 internal drop = 1.95 FT; DMH-1 external drop = 3.95 FT.

Storm/sanitary modeled vertical clearance = 1.252694 FT (reviewed assumption).

PDF / GeoPackage / semantic package share stable IDs and geometry within 0.01 FT.

Public capacity, HGL, tie/outfall approval, survey, infiltration, and compliance remain UNKNOWN.

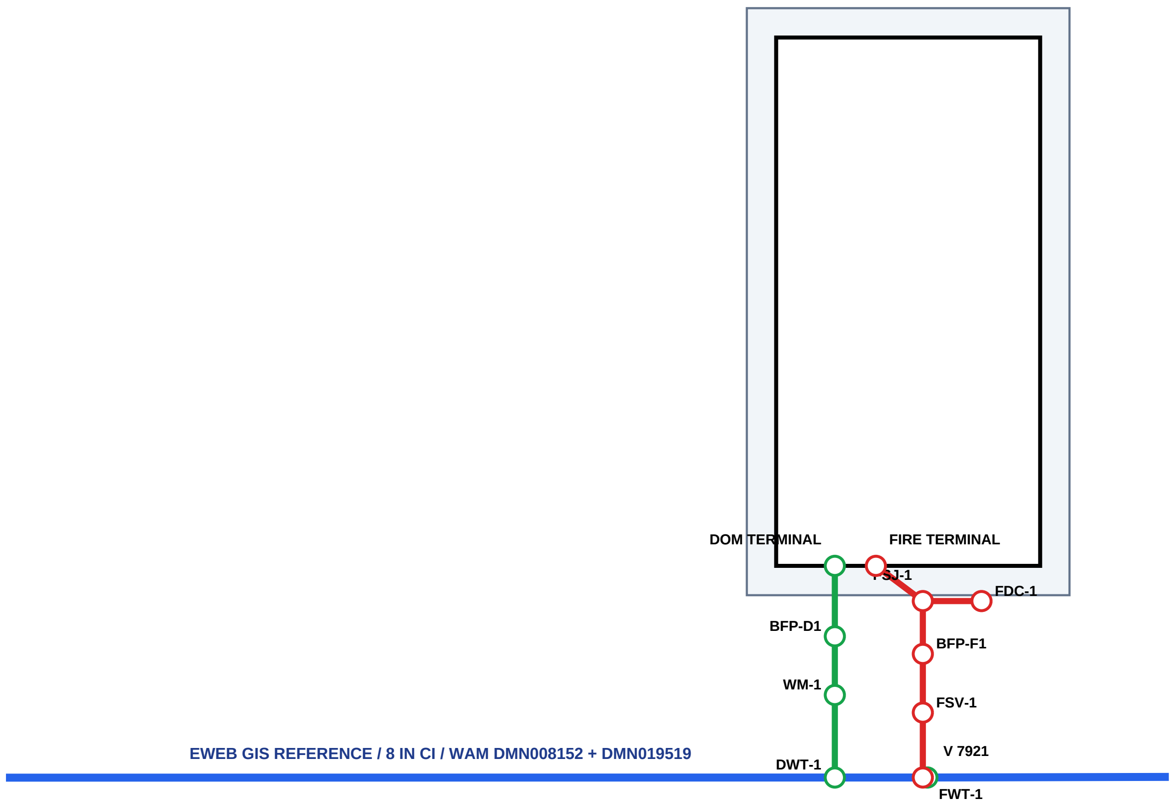
FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

Conceptual test detail only; no professional engineering or permit-compliance claim

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

MODEL STUDIO - HILYARD APARTMENT CIVIL PLAN TEST DOMESTIC WATER / FIRE SERVICE PLAN

Separate fictional pressure networks from reference GIS context; no pressure, flow, capacity, hydrant-test, tie, meter, backflow, or Fire Marshal approval claim



REFERENCE / DESIGN BASIS

Fronting context: active 8 IN CI EWEB GIS main

GIS reference only - NOT SURVEY OR TIE AUTHORITY

Domestic: 2 IN HDPE DR11 / master meter / RPBA

Fire: 6 IN C900 DR18 / valve / detector double check

FDC branch: 4 IN assumption to FSJ-1

Modeled cover: 3.50 FT / declared minimum 3.00 FT

Pipe elevations and final grades: UNKNOWN

Pressure, residual pressure, capacity: UNKNOWN

Hydrant flow test and hydraulic model: UNKNOWN

Meter/backflow/tie/Fire Marshal approval: UNKNOWN

PLAN-SEPARATION SCREEN

Domestic to sanitary: 62.80 FT / 10.00 FT minimum

Domestic to storm: 39.36 FT / 2.00 FT minimum

Fire to sanitary: 64.29 FT / 10.00 FT minimum

Fire to storm: 33.29 FT / 2.00 FT minimum

All values are horizontal plan distance only

Vertical separation: UNKNOWN - pothole/survey required

No water line crosses a modeled storm facility

FIELD HOLD POINTS

1 Obtain EWEB pressure/capacity and approved tie basis
2 Calculate building demand and sprinkler/fire flow
3 Survey/locate/pothole utilities and final grades
4 Approve meter, backflow, FDC, valves, and restraint
5 Inspect tracer/bedding/restraint; pressure/disinfect/test
6 Capture certified tests and as-built coordinates/depths

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

Pressure-network plan only; vertical profile omitted because pipe elevations/final grades are unknown

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

MODEL STUDIO - HILYARD APARTMENT CIVIL PLAN TEST WATER / FIRE SEPARATION SCHEDULE / TEST DETA

Stable pressure-network IDs, horizontal coordination checks, restraint notes, field workflow, and explicit hydraulic/approval unknowns

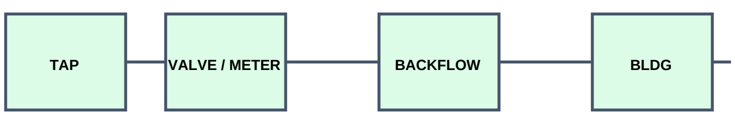
PRESSURE SERVICE EDGE SCHEDULE

EDGE ID	TYPE	DIA	LEN	COVER	MATERIAL	RESTRAINT	HYDRAULICS
dom-edge-tie-meter	private_domestic_service	2 IN	14.06	3.50 FT	HDPE DR11	manufacturer fused/re...	UNKNOWN
dom-edge-meter-backflow	private_domestic_service	2 IN	10.00	3.50 FT	HDPE DR11	fused service; device...	UNKNOWN
dom-edge-backflow-terminal	private_domestic_service	2 IN	12.00	3.50 FT	HDPE DR11	fused service with re...	UNKNOWN
fire-edge-tie-valve	private_fire_service	6 IN	11.06	3.50 FT	C900 PVC DR18	restrained joint/thru...	UNKNOWN
fire-edge-valve-backflow	private_fire_service	6 IN	10.00	3.50 FT	C900 PVC DR18	restrained joints at ...	UNKNOWN
fire-edge-backflow-junction	private_fire_service	6 IN	9.00	3.50 FT	C900 PVC DR18	restrained tee approa...	UNKNOWN
fire-edge-junction-terminal	private_fire_service	6 IN	10.00	3.50 FT	C900 PVC DR18	restrained tee and bu...	UNKNOWN
fire-edge-fdc-branch	fdc_branch	4 IN	10.00	3.50 FT	C900 PVC DR18	restrained FDC check-...	UNKNOWN

HORIZONTAL SEPARATION / VERTICAL-STATUS SCHEDULE

RELATIONSHIP ID	SYSTEMS	PLAN CLEAR	MINIMUM	VERTICAL	STATUS
domestic-sanitary-separation-01	domestic_water / sanitary	62.80 FT	10.00 FT	UNKNOWN	PLAN PASS / FIELD VERIFY
domestic-storm-separation-01	domestic_water / storm	39.36 FT	2.00 FT	UNKNOWN	PLAN PASS / FIELD VERIFY
fire-sanitary-separation-01	fire_water / sanitary	64.29 FT	10.00 FT	UNKNOWN	PLAN PASS / FIELD VERIFY
fire-storm-separation-01	fire_water / storm	33.29 FT	2.00 FT	UNKNOWN	PLAN PASS / FIELD VERIFY

CONCEPTUAL DEVICE / RESTRAINT DETAIL - NOT A CONSTRUCTION DETAIL



Continuous tracer/warning system; bedding/backfill/compaction per accepted details.

Restrain valves, tees, bends, device transitions, and building entry from approved thrust analysis.

Exact devices, vault/box drainage, freeze protection, and final grade/elevation remain UNKNOWN.

UNRESOLVED DESIGN / APPROVAL GATES

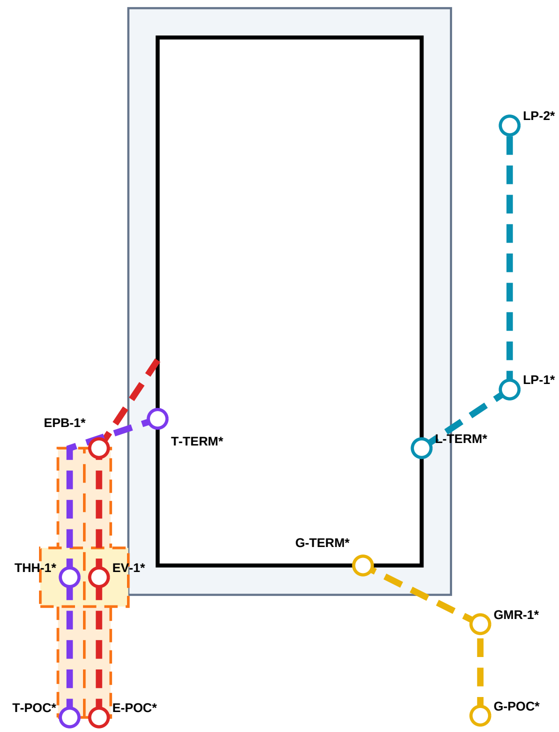
1. Building occupancy, fixture demand, irrigation/common-use demand
2. Required fire flow, sprinkler demand, hose allowance, duration
3. EWEB static/residual pressure, capacity, system impact, tie location
4. Hydrant flow test and water/fire hydraulic calculations
5. Meter and backflow hazard/device/placement approval
6. FDC/riser/access/signage and Fire Marshal approval
7. Pipe centerline elevations, final grades, crossings, thrust/restraint
8. Survey, permits, testing, disinfection, inspection, and acceptance

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

MODEL STUDIO - HILYARD APARTMENT CIVIL PLAN TEST DRY UTILITIES / JOINT TRENCH PLAN

ALL PROPOSED DRY ROUTES ARE DASHED REVIEWED ASSUMPTIONS - NOT LOCATED UTILITIES OR OWNER-APPROVED DESIGNS



OWNER / PROVENANCE BASIS

Electric owner: EWEB (confirmed jurisdiction)

Gas owner: NW Natural (confirmed Eugene service)

Telecom provider at parcel: UNKNOWN

Every route/POC/box: REVIEWED ASSUMPTION

Dashed line = NOT LOCATED / NOT APPROVED

Power + telecom share a coordination trench

EV-1 and THH-1 are separate owner enclosures

Gas uses a separate fictional route

Site lighting is a separate building-fed branch

UNRESOLVED DESIGN GATES

Electric load, voltage, transformer and capacity

EWEB point of service, duct bank, vault/box design

Telecom provider, capacity, POC, boxes and bends

Gas main/location, pressure, capacity, size and meter

Depths, separations, crossings and final grades

Lighting circuit, fixtures, controls and photometrics

Survey/locates, owner designs, permits and acceptance

FIELD SAFETY HOLD POINTS

1 Call/coordinate utility locates before excavation

2 Obtain owner-issued plans and approved service points

3 Pothole/survey crossings and reconcile vertical data

4 Inspect conduit/pipe, boxes, tracer and trench open

5 Capture tests and owner-accepted as-built records

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

Dashed assumed routes only; no utility locate, capacity, owner design, or construction authorization

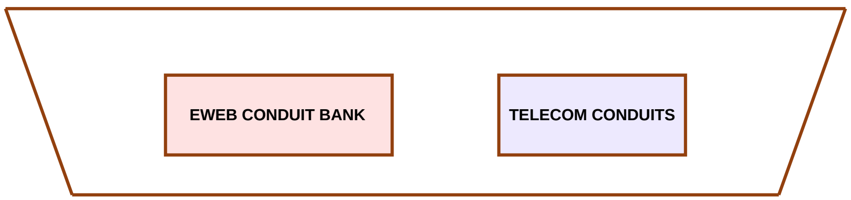
MODEL STUDIO - HILYARD APARTMENT CIVIL PLAN TEST DRY UTILITY SCHEDULE / COORDINATION DETAILS

Stable network IDs, joint-trench relationships, owner gates, field workflow, and explicit unresolved service inputs

DRY UTILITY EDGE SCHEDULE

EDGE ID	SYSTEM	COUNT	SIZE	LEN	MATERIAL	TRENCH	STATUS
gas-edge-poc-meter	gas_service	1	2 IN	15.60	polyethylene gas servi	separate_route	ASSUMED
gas-edge-meter-terminal	gas_service	1	2 IN	22.36	polyethylene gas servi	separate_route	ASSUMED
power-edge-poc-vault	electric_conduit_bank	3	4 IN	23.90	PVC conduit - exact sc	joint_use_power_te	ASSUMED
power-edge-vault-pull	electric_conduit_bank	3	4 IN	22.00	PVC conduit - exact sc	joint_use_power_te	ASSUMED
power-edge-pull-terminal	electric_service_conduit	3	4 IN	18.03	PVC conduit - exact sc	electric_service_b	ASSUMED
telecom-edge-poc-handhole	telecom_conduit_bank	2	2 IN	23.90	PVC communications con	joint_use_power_te	ASSUMED
telecom-edge-handhole-terminal	telecom_service_conduit	2	2 IN	37.81	PVC communications con	joint_use_then_ser	ASSUMED
lighting-edge-terminal-pole-01	site_lighting_conduit	1	2 IN	18.03	PVC electrical conduit	site_lighting_bran	ASSUMED
lighting-edge-pole-01-pole-02	site_lighting_conduit	1	2 IN	45.00	PVC electrical conduit	site_lighting_bran	ASSUMED

CONCEPTUAL JOINT-TRENCH COORDINATION - NOT A CONSTRUCTION SECTION



Actual depth, horizontal/vertical separation, warning systems, bedding, and backfill are OWNER-DESIGNED UNKNOWNs.

EV-1 and THH-1 are separate enclosures inside a graphic coordination zone; no shared physical vault is claimed.

OWNER / REVIEW MATRIX

EWEB: confirmed electric owner; route/load/design UNKNOWN

NW Natural: confirmed Eugene gas utility; parcel main/tie UNKNOWN

Telecom: Eugene providers exist; serving provider UNKNOWN

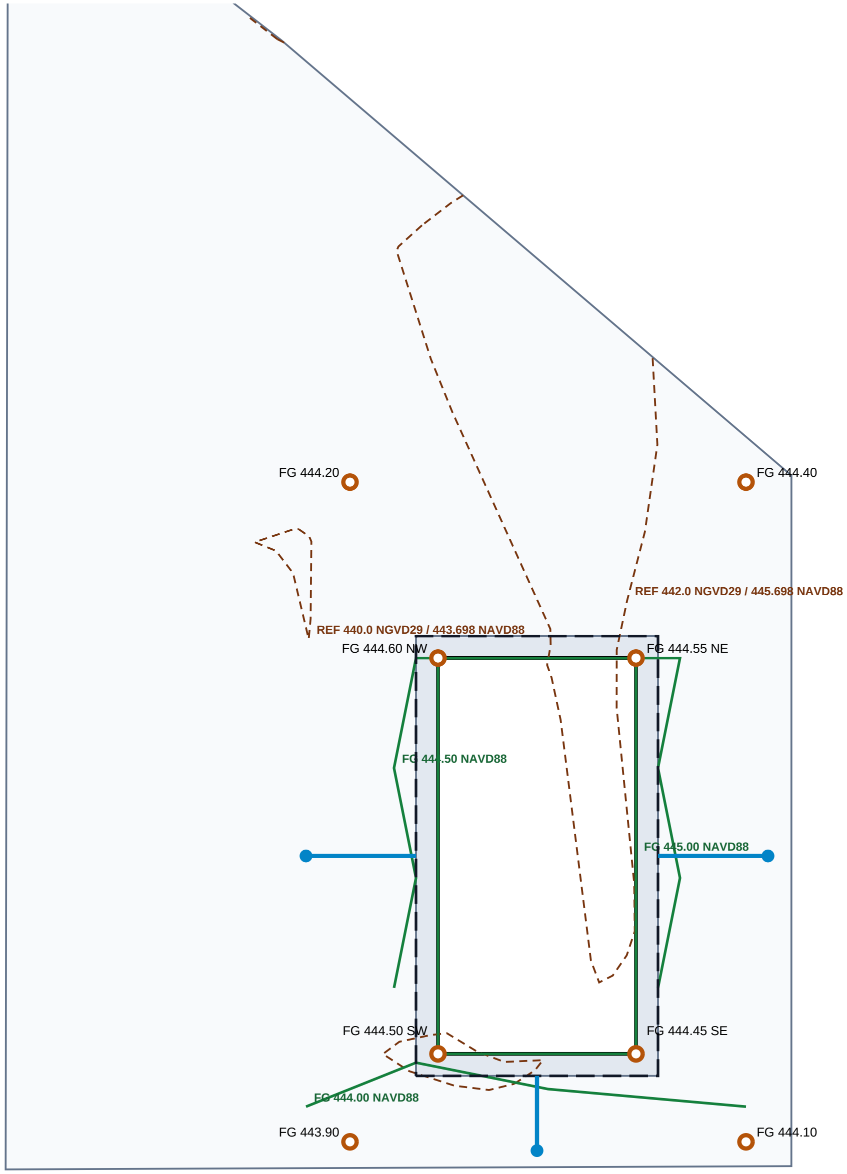
Site lighting: private design; electrical/photometric basis UNKNOWN

All routes: REVIEWED ASSUMPTION / DASHED / NOT LOCATED

All capacities and owner approvals: UNKNOWN

MODEL STUDIO - HILYARD APARTMENT CIVIL PLAN TEST GRADING / DRAINAGE PLAN

City 1999 reference contours explicitly converted NGVD29 to NAVD88; proposed grading is a fictional reviewed assumption



VERTICAL-DATUM CONTROL

Source: City of Eugene 2-FT contours / 1999 orthophotos

Source datum: NGVD29 / source geometry EPSG:2914

Canonical datum: NAVD88 / geometry EPSG:6823

NOAA VDatum / VERTCON 3.0 shift: +3.698 FT

Reported conversion uncertainty: 0.165 FT

Shift range across parcel: 0.004 FT

Original and converted elevations retained on every contour

REFERENCE-GRADE ONLY / NOT A TOPOGRAPHIC SURVEY

PROPOSED-GRADE BASIS

Building FFE: 445.00 FT NAVD88 (reviewed assumption)

Corner FG: 444.45-444.60 FT NAVD88

Modeled FFE freeboard: 0.40-0.55 FT

Contours/spots/arrows derive from one canonical model

Drainage capacity and overflow route: UNKNOWN

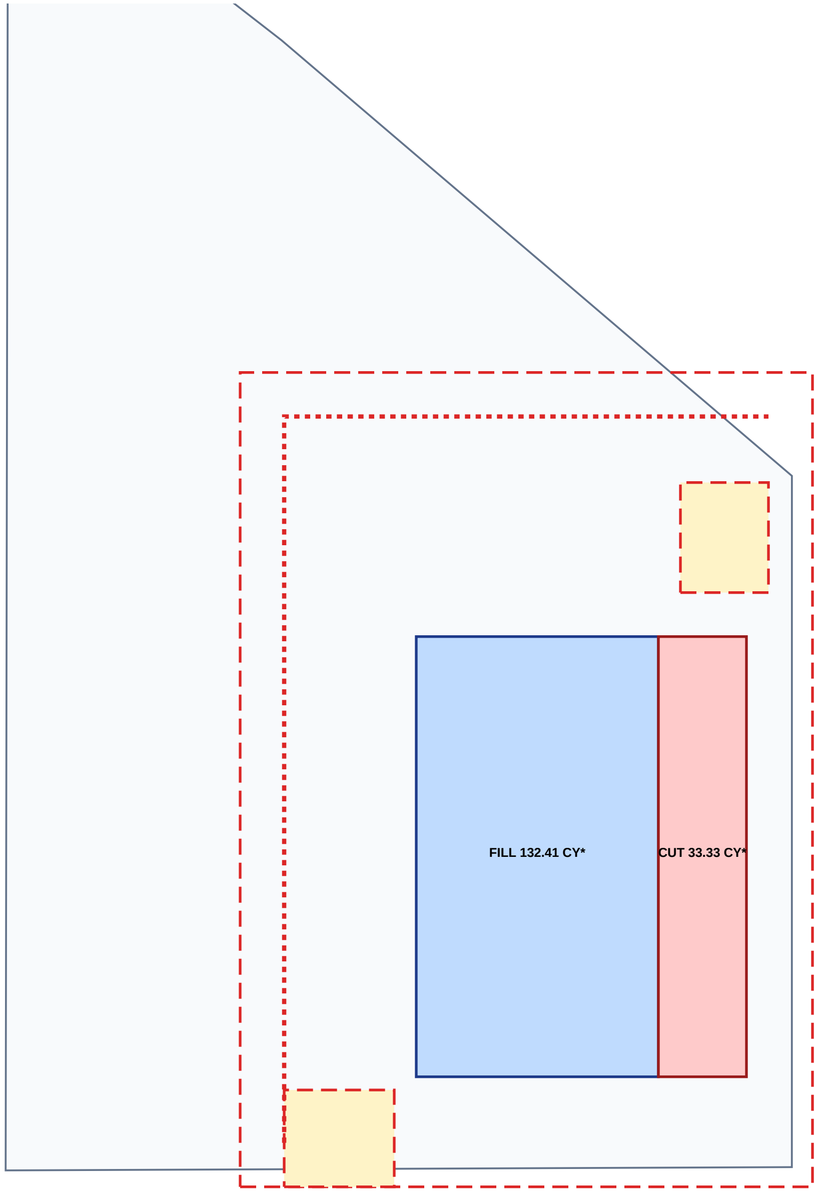
ADA, pavement, curb, landscape and final grading: UNKNOWN

Supersede with current survey and licensed final design

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

MODEL STUDIO - HILYARD APARTMENT CIVIL PLAN TEST EARTHWORK / SITE PREPARATION PLAN

Screening-only cut/fill and temporary construction geometry; no survey-to-surface volumes or approved erosion-control plan



SCREENING EARTHWORK SUMMARY

FILL: 5500 SF x 0.65 FT / 27 = 132.41 CY

CUT: 2000 SF x 0.45 FT / 27 = 33.33 CY

NET IMPORT: 99.07 CY

Shrink/swell, topsoil, unsuitable soil and waste: UNKNOWN

No TIN-to-TIN survey volume calculation was performed

QUANTITIES ARE NOT FOR BID OR CONSTRUCTION

TEMPORARY SITE-PREP FEATURES

Disturbance limit: reviewed fictional assumption

Construction entrance: location/section unapproved

Stockpile: stabilization and capacity unresolved

Silt fence: assumed alignment; no approved ESC design

Construction sequence and inspection plan: UNKNOWN

All temporary assets remain in the temporary phase

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

Screening quantities and temporary assumptions only; not an approved construction or erosion-control plan